MIL-R-24061B(EC)
10 July 1972
SUPERSEDING
MIL-R-24061A(EC)
6 March 1970
(See 6.3)

MILITARY SPECIFICATION

RADIAC SET, AN/PDR-27( )

1. SCOPE

1.1 This specification covers Radiac Set AN/PDR-27( ), lightweight, shipboard, portable equipment for detecting, indicating, and computing gamma radiation, and for indicating presence of beta radiation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	- Box, Fiberboard
PPP-T-60	- Tape, Packaging waterproof
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing)

MILITARY

MIL-E-1/906	- Electron Tube, Radiation Counter, Type 5979
MIL-E-1/973	- Electron Tube, Radiation Counter, Type 5980
MIL-B-18	- Battery, Dry
MIL-P-116	- Preservation, Methods of
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-D-3464	- Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-M-10304	- Meter, Electrical Indicating, Panel Type, Ruggedized General Specification for
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore, General Specification
MIL-E-17555	- Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of
MIL-M-19590	- Marking of Commodities and Containers to Indicate Radioactive Material
MIL-P-24052	- Preliminary Plans and Test Procedure Requirements for Radiac Instruments
MIL-I-45208	- Inspection System Requirements

FSC 6665

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108 - Definition of and Basic Requirement for Enclosures for Electric and Electronic Equipment
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" X 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781 - Reliability Tests: Exponential Distribution
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

DRAWINGS

MILITARY

RE-D2681324 - Cord, Probe, Retractile
RE-F2688037 - H-43B/U Headset Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Ill. 60606).

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Rules

(Application for copies should be addressed to National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, D. C. 20036).

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DEPARTMENT OF TRANSPORTATION

Code of Federal Regulations, Titles 10 and 49.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2). Preliminary plans and test procedures shall be in accordance with MIL-P-24052 (see 6.1.2).

3.2 General. The equipment shall be in accordance with MIL-E-16400 except as hereinafter specified.

3.3 Reliability.

3.3.1 Quantitative reliability requirement. The specified mean time between failure (MTBF) (θ_o as defined by MIL-STD-781) of this equipment shall be 3000 hours.

3.3.2 Reliability program plan. The contractor shall submit a reliability program plan in accordance with and in paragraph sequence with MIL-STD-785. Particular attention should be paid to those sections of MIL-STD-785 which require submission of detailed tasks, procedures to implement and control these tasks, and time schedule. The proposed reliability program plan shall be submitted as a separate and complete entity within the total system or equipment project delineation. The reliability program plan as approved and incorporated into the contract becomes the basis for contractual compliance. The plan, including reliability test plans, must describe an integrated effort within the total project; it shall provide information to show how the contractor will meet specified quantitative reliability requirements during development and manufacture. The final reliability program plan shall be submitted for Government approval (see 6.1.2).

3.3.3 Reliability prediction. The contractor shall submit a reliability prediction in accordance with MIL-HDBK-217 to the command or agency concerned. The prediction shall assume a 50°C temperature and a shipboard environment. The contractor shall evaluate the prediction with respect to the reliability expected from a production unit. If there is a discrepancy between the predicted MTBF and required MTBF, the contractor shall identify the causes of the discrepancies and indicate action required to ensure that the inherent reliability of the equipment is preserved during the production process. The processes and procedures required to ensure production of reliable equipment shall be thoroughly documented (see 6.1.2).

3.3.4 Source of failure rates. The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned. When such alternate sources of data are proposed, enough information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.

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3.3.5 Reliability reports. Reports providing a complete accounting of progress on each element defined by the reliability program plan, results achieved, and status of actions to resolve problems and correct weak links shall be submitted to the command or agency concerned (see 6.1.2).

3.4 Design requirements.

3.4.1 Construction (modular). The requirements for modular construction in MIL-E-16400 are applicable.

3.4.1.1 Equipment composition. The Radiac Set AN/PDR-27( ) shall consist of the following:

(a) One radiacmeter, including probe, probe cable, and form-fitting handle, but not including batteries.
(b) One high-impedance headset, Type H-43B/U, in accordance with drawing RE-F2688037.
(c) One radioactive test sample.
(d) One carrying strap.
(e) One spare tube type 5979 in accordance with MIL-E-1/906.
(f) One spare tube type 5980 in accordance with MIL-E-1/973.
(g) One spare voltage-regulator tube (see 3.4.8.7).
(h) One technical manual.
(i) One carrying case.

3.4.2 Environmental conditions. The equipment shall be capable of operation under the following conditions as specified in MIL-E-16400 or as modified below:

(a) Temperature, class 3.
(b) Humidity.

3.4.3 Electrical design. The nominal supply voltage shall be not more than 9 volts direct current (Vdc). The power source shall be batteries, Type BA-30 (D-cells), in accordance with MIL-B-18. Batteries are not required to be furnished with the set.

3.4.3.1 External connectors. External connectors shall be in accordance with MIL-E-16400.

3.4.4 Maintenance features.

3.4.4.1 Test points. Test points shall be identified, but shall not be brought out to the case.

3.4.4.2 Blown fuse indicators. Blown fuse indicators are not required.

3.4.5 Shock. The equipment shall be capable of withstanding the grade A, class I, lightweight, type A shock requirements in accordance with MIL-S-901.

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3.4.6 Controls, indicators, and panel layout.

3.4.6.1 Controls. All controls shall be classified as continually required, except calibration controls.

3.4.6.2 Elapsed time meters. Elapsed time meters are not required.

3.4.7 Technical information. Instruction plates are not required, but required technical information shall be in the technical manual.

3.4.8 Performance and special requirements.

3.4.8.1 Radiacmeter. The radiacmeter shall be capable of detecting and measuring gamma radiation in the intensity range up to 500 milliroentgens per hour (mr/hr) subject to further requirements herein, and be capable of detecting the presence of beta radiation.

3.4.8.1.1 Radiacmeter composition. The radiacmeter shall include the following subassemblies and circuitry:

(a) Stable electronic pulse rate computer and integrating circuit.
(b) Detector probe with connecting cable.
(c) One multi-range indicating meter assembly.
(d) One control switch and range selector assembly.
(e) One headset connection and dust-tight cover.
(f) One lamp for night illumination, with control by switch only.
(g) Housing to contain and support the above and associated components.

3.4.8.1.2 Radiacmeter operation. The radiacmeter operation shall be as follows:

(a) Accuracy. Within the environmental limits specified herein, when the radiacmeter is exposed to gamma radiation whose true value is between 10 and 100 percent of the scale range selected, the reading shall not deviate more than ±20 percent from the true value. For this determination, the radiation shall be incident normal to the plane formed by the major axes of two Geiger-Muller tubes (see 3.4.8.3).
(b) Calibration. Calibration shall be performed with a gamma source or a measuring instrument which shall have been calibrated by the National Bureau of Standards. Calibration equipment and procedure shall be approved in advance by the command or agency concerned.
(c) Calibration controls. A separate control shall be provided for calibration of each scale range. Access to the controls shall be restricted to prevent tampering.
(d) Saturation. When the radiacmeter is exposed to gamma radiation up to 600 roentgens per hour, it shall not saturate. Whatever gamma measurement scale range is selected, the meter pointer shall indicate off-scale at the high end of the scale.

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(e) Power supply. A power supply shall be furnished as part of the radiacmeter to provide all necessary voltages for operation of the detector, computer integrator indicator, and headset, employing the batteries as prime source. The power consumption shall be limited to a value such that the radiacmeter will operate continuously for 40 hours or more under normal operating conditions within the stated accuracy limits on one set of batteries. Battery end point shall be 1.1 volts per cell. The 40 hours include 8 hours with illuminating lamp ON.

(f) Energy response. The response to gamma radiation at any given intensity shall vary not more than ±20 percent throughout the energy range from 0.08 million electron volts (MeV) to 1.33 MeV. Below 0.08 MeV, the relative response shall decrease with decreasing energies (see 4.3.3).

3.4.8.1.3 Pulse rate computer integrator function. The computer integrator shall receive pulses from the detector at the input and shall furnish at the output an electrical current which shall be directly proportional to the average value of the rate of arrival of pulses, but independent of their amplitude and duration. The computer integrator shall drive a 50-microampere indicating meter. An additional output shall present signals representing individual pulses to drive a headset.

3.4.8.1.4 Detector-computer-integrator response. The radiacmeter shall measure the intensity of gamma radiation between zero and 500 milliroentgens per hour under the following conditions:

(a) Circuit time constant. For an output current sufficient to produce a deflection of 20 percent or more of full-scale reading, the circuit time constants shall be such that the meter pointer shall reach 90 percent of true indication in not less than 3 seconds and not more than 7 seconds.

(b) Noise. No noise or spurious counts shall be introduced, except from operation of the selector switch. This necessary zero background shall be provided as the result of proper care in design, construction, and choice of parts.

3.4.8.1.4.1 Detector-computer-integrator circuit. The circuit shall include no electron tubes except the Geiger-Muller tubes in the detector.

3.4.8.1.5 Operating controls. The operating controls shall consist of a selector switch and a switch to operate the illuminating lamp.

3.4.8.1.5.1 Selector switch. The control for the selector switch shall be located on the top of the radiacmeter housing and shall be rotatable to six positions to provide for the modes of operation listed. This switch shall be coupled to the meter so that the scale corresponding to the switch position is presented.

(a) Position 1 - OFF. Maximum counterclockwise. The switch shall open at least the minimum number of circuits to ensure zero current drain from the battery. Meter scale: OFF - BATT (Same as for (b)).

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(b) Position 2 - BATTERY CONDITION. The switch shall connect the meter into the circuit in such a way as to indicate the condition of the battery under load equal to the circuit load. Meter scale: OFF - BATT (Same as for (a)).
(c) Position 3 - 0-500 milliroentgens per hour. Meter scale: 0 to 500; Detector tube in circuit: type 5980 only.
(d) Position 4 - 0-50 milliroentgens per hour. Meter Scale: 0 to 50; Detector tube in circuit: type 5980 only.
(e) Position 5 - 0-5 milliroentgens per hour. Meter scale: 0 to 5; Detector tubes in circuit: 5980 and 5979 (both) (see 3.4.1.1).
(f) Position 6 - 0-0.5 milliroentgens per hour. Meter scale: 0 to 0.5; Detector tubes in circuit: 5980 and 5979 (both) (see 3.4.1.1).

3.4.8.1.5.2 Illumination. A push-to-operate switch shall be included on the top of the panel to control the illumination lamp. The switch shall be located adjacent to the carrying handle so that the operator may hold the radiacmeter by the handle and operate the switch with the same hand. The lamp shall illuminate the meter sufficiently to permit reading-after-dark adaptation. The lamp shall be an incandescent type, with optics, if necessary, to direct light where required.

3.4.8.1.6 Indicating meter. The indicating meter shall be a 5-position ruggedized multi-range scale shift microammeter, 50 microamps full scale, type MR36M201 SPECR in accordance with MIL-M-10304. The meter shall be mechanically coupled to the selector switch so that the scale positions and switch positions will correspond. The first position of the meter shall be common to Positions 1 and 2 of the selector switch. The meter shall be located at the top of the radiacmeter housing so as to be visible from above the radiacmeter through a window, whether the radiacmeter is carried by the hand, by the handle, or is carried in the abdominal or hip position by the strap. The meter shall be sealed to the housing to maintain the watertight degree of enclosure.

3.4.8.2 Radiacmeter housing. The radiacmeter housing shall be a watertight enclosure to contain the electronic circuitry (except the probe), meter, and batteries. The material of the housing shall be aluminum.

3.4.8.2.1 Radiacmeter housing construction. The housing shall be of split-type gasketed construction, with the seam at the top, and held closed by captive screws inserted from the top surface. The top shall act as a panel to support the indicating meter and the electronic chassis. The top shall also include an integral compartment for the batteries. This compartment shall be accessible only through a separate cover in the upper surface, held by captive screws. The top shall be fitted to receive the carrying strap and the carrying handle. The battery connectors and insulator shall be impervious to battery fluids. The bottom of the housing shall be a cover which shall enclose the electronic chassis and the battery compartment.

3.4.8.2.2 Probe well. On the back end of the exterior of the housing there shall be a receptacle for holding the probe in a vertical position. The receptacle shall be integral with the lower part of the housing.

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3.4.8.2.3 Radiacmeter finish. Unless otherwise specified in the contract or order (see 6.1.1), the radiacmeter housing shall be treated against corrosion and painted gray in accordance with MIL-E-16400. The surface shall be as smooth as possible with the minimum number of crevices to collect contaminated material. Rubber or felt feet shall not be permitted. The surface shall be decontaminable easily without recourse to laboratory methods.

3.4.8.2.4 Hand carrying handle. A form fitting hand-carrying handle shall be attached to the top of the radiacmeter housing. The handle shall be recessed to store the probe cable in the coiled condition. The center of gravity of the radiacmeter and the location of the handle shall be adjusted to provide for proper balance.

3.4.8.2.5 Calibration access cover. A cover shall be provided in the top panel to protect the calibration controls. The cover shall not compromise the watertightness of the enclosure.

3.4.8.3 Detector (probe). The radiacmeter shall be equipped with a detector (probe) which shall house two Geiger-Muller tubes, types 5980 and 5979, covering ranges of 0 to 500 mr/hr and 0 to 5 mr/hr, respectively. The probe shall consist of two smooth-contoured cylindrical metal housings, one for each tube, held together by clamps, with their major axes and the major axes of the tubes parallel. There shall be a movable beta-ray shield opposite the beta window of the type 5979. The overall outside diameter of the housing for the type 5979 shall be not more than 1.350 inches for a length of 0.375 inch from the end with the beta shield, and the same end of the housing (not including the beta shield) shall project not more than 0.178 inch beyond the end of the tube.

3.4.8.3.1 Detector shielding. The detector shielding shall be so mounted as to permit ready access to and replacement of the tubes.

3.4.8.3.2 Detector cable. The cable between the radiacmeter housing and the probe shall be retractile type in accordance with drawing RE-D2681324. Strain relief devices shall be provided at the points of attachment to the probe and radiacmeter.

3.4.8.3.3 Degree of enclosure. The degree of enclosure of the probe shall be watertight in accordance with MIL-STD-108.

3.4.8.4 Headset assembly. One headset, high-impedance, type H-43B/U, shall be included in the Radiac Set AN/PDR-27( ). It shall be connected to the headset receptacle when in use, and stored in the carrying case when not in use.

3.4.8.5 Carrying strap. There shall be one carrying strap, or shoulder harness, included in the Radiac Set AN/PDR-27( ). It shall provide for carrying the radiacmeter in either the hip or abdominal position, and shall be attachable to the designed points on the top of the radiacmeter housing. When not in use, it shall be stored in the carrying case.

3.4.8.6 Radioactive test sample. A gamma-emitting radioactive test sample shall be provided to test qualitatively the overall functioning of the radiacmeter. It shall consist of an aluminum rod enclosing at one end a capsule of

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Krypton-85 of not more than 5 millicuries. The test sample shall meet the requirements of the Code of Federal Regulations, Title 10, Part 30, and shall be subject to approval of the command or agency concerned. When not is use, the test sample shall be stored in the carrying case.

3.4.8.7 Spare tubes. The following spare tubes shall be provided and stored in the carrying case:

(a) One type 5979 in accordance with MIL-E-1/906.
(b) One type 5980 in accordance with MIL-E-1/973.
(c) One high-voltage regulator tube, if required in high voltage circuit.

3.4.8.8 Carrying case. The Radiac Set AN/PDR-27( ) shall include one carrying case, adequately compartmented, to store and protect the following components:

(a) Radiacmeter.
(b) Headset H-43B/U.
(c) Radioactive test sample.
(d) Carrying strap (harness).
(e) One each spare tubes, type 5979, 5980, and also high voltage regulator (see 3.4.8.7).
(f) One technical manual.
(g) One set of spare batteries.

3.4.8.8.1 Marking. The carrying case shall be marked to indicate the presence of radioactive material in accordance with the requirements of MIL-M-19590. The marking shall be placed on the same side as the identification plates.

3.4.8.8.2 Material. The material of the carrying case shall be drawn aluminum, as light as possible consistent with the strength required.

3.4.8.8.3 Degree of enclosure. The degree of enclosure shall be splash-proof in accordance with MIL-STD-108.

3.4.8.9 Altitude. The radiacmeter shall be designed to meet the altitude test specified in 4.3.2.

3.4.8.10 Radiac set, size and weight. The size and weight of the following components shall be no greater than specified herein:

(a) Radiacmeter. Length - 11-17/32 inches; Width - 5 inches; Height - 8-1/2 inches; Weight - 10.5 lbs.
(b) Probe. Length - 7-5/8 inches; Width - 2-9/16 inches; Height - 1-19/32 inches; Weight - 1.1 lbs.
(c) Test sample. Length - 5 inches; Diameter 3/8 inch.

3.4.8.10.1 Carrying case. The carrying case shall be provided with at least one handle for transporting purposes.

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3.4.8.11 Electromagnetic interference (EMI). The equipment shall be class 1D of MIL-STD-461 and meet the requirements of RE02, RS01, and RS03.

3.5 Maintainability. The equipment shall be constructed to provide ease of maintenance, accessibility, and replacement of all modules and parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system requirement. The contractor shall provide and maintain a written inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.1.1.1 Quality assurance plan. A quality assurance plan applicable to the equipments covered by this specification shall be prepared for Government approval (see 6.1.2).

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to ensure compliance with all the requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. The test procedures shall be prepared in accordance with MIL-P-24052 and as specified herein. The contractor shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and production inspection to the command or agency concerned, via the cognizant Government inspector, for review and approval prior to beginning test. Each test shall be identified in accordance with the classifications of 4.2 (see 6.1.2).

4.2 Classification of inspection. The methods of examination and testing of the equipment are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

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(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)
(3) Environmental inspection (see 4.4.3)
(4) Reliability demonstration (see 4.4.4)

(c) Inspection of preparation for delivery (see 4.5)

4.3 First article inspection. Unless otherwise specified (see 6.1.1), three units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in 4.3.1 and MIL-E-16400 tests as follows:

(a) Surface examination
(b) Operating test
(c) Weights and dimensions
(d) Enclosure
(e) Temperature
(f) Humidity
(g) Electromagnetic interference (RE02, RS01, 2nd RS03 of MIL-STD-462
(h) Controls
(i) Shock, vibration, and inclination
(j) Salt spray
(k) Magnetic

4.3.1 In addition to the tests and inspections of 4.3, the following tests shall be performed to determine conformance with this specification:

(a) Stability (see 3.4.8.1.2(e))
(b) Accuracy, at 1/4, 1/2, and 3/4 points on each scale range (see 3.4.8.1.2(a))
(c) Altitude (see 4.3.2)
(d) Saturation (see 3.4.8.1.2(d))
(e) Energy response (see 3.4.8.1.2(f) and 4.3.3)
(f) Power test (battery life) (see 3.4.8.1.2(e) and 4.3.4)
(g) Reliability (see 4.4.4)

4.3.2 Altitude test. The equipment shall actively monitor a source (which shall produce a midscale reading on the 0 to 5 mr/hr scale) for a period of 24 hours while exposed to an air pressure of 8.8 inches of mercury at a temperature of -40°C, and shall then be turned off and exposed to an air pressure of 3.4 inches of mercury at a temperature of -54°C for an additional 23-hour period. The equipment shall conform to the performance requirements of this specification throughout the first 24-hour period and subsequent to this test.

4.3.3 Energy response test. The energy response of the radiacmeter shall be determined by irradiation with gamma radiation from Cobalt-60 (or with Cesium-137, with appropriate correction factors), and with X-Rays at approximately 200, 160, 120, 80, 50 and 30 thousand electron volts (keV). Tube voltages and filtrations for the particular energies shall be as specified by the National Bureau of Standards for a 250 kilovolts peak (kVP) machine. The response shall fall within the limits specified in 3.4.8.1.2(f).

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4.3.4 Power test. This test shall be conducted with the batteries specified in 3.4.3 to demonstrate the requirements of 3.4.8.1.2(e). Accuracy shall be within the limits specified in 3.4.8.1.2(a) at the completion of this test. Current drain shall be established for use as a production test measurement.

4.3.5 First article inspection report. A first article inspection report shall be prepared containing the results of the first article inspection. The report shall include completed form NAVEXOS 3373 as specified in MIL-S-901 for the shock test and resilient mount data sheet as specified in MIL-E-16400 (see 6.1.2).

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the following:

(a) Surface examination (see 4.3(a))
(b) Operating test (see 4.3(b))
(c) Stability (see 4.3.1(a))
(d) Accuracy (see 4.3.1(b))
(e) Power (current drain) (see 4.3.4)

4.4.2 Production control inspection. Production control inspection shall be conducted on a sampling basis in accordance with 4.4.2.1. This inspection shall comprise such examination and testing as will prove satisfactory performance through the entire range of operation, detect deterioration of the design or deviations from approved processing of materials, and demonstrate that the equipment will function satisfactorily under adverse conditions. Unless otherwise specified herein, these tests shall be conducted at normal ambient room temperature. This inspection shall include the following tests:

(a) Enclosure test (see 4.3(d))
(b) Power (battery life) (40 hour minimum) (see 4.3.1(f))
(c) Controls (see 4.3(h))
(d) Saturation (see 4.3.1(d))
(e) Reliability (see 4.4.4)

4.4.2.1 Sampling plan. From each lot which has passed production inspection (see 4.4.1), a random sample of units shall be selected and subjected to production control inspection. Sampling shall be in accordance with MIL-STD-105, General Inspection Level I, and an acceptable quality level (AQL) of 4.0 percent.

4.4.2.2 Rejection. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate

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from the new lots and shall not lose their identity. The results of each test shall be compared with the requirements of this specification. Failure to conform to this specification for any test shall be counted as a defect, and the unit shall be rejected. If the number of rejections in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4.3 Environmental inspection. Environmental inspection shall be conducted on a sampling plan as specified in 4.4.3.1, except that the Government may waive this requirement in the light of evidence that the contractor's production and quality control are such as to ensure manufacture of satisfactory equipment. These tests shall include the following:

(a) Temperature (see 4.3(e))
(b) Humidity (see 4.3(f))
(c) Shock (see 4.3(i))
(d) Vibration (see 4.3(i))
(e) Electromagnetic interference (see 4.3(g))

4.4.3.1 Sampling plan. From each lot which has passed the production inspection, a random sample of units shall be selected for environmental tests. These tests may be conducted concurrently with production control inspection of different samples from the lot being tested. The sample size and lot acceptance level shall be based upon table I. The Government inspector may at his discretion, further reduce sample sizes.

TABLE 1. Sampling for environmental inspection.

Lot size	Sample size	Number of units nonconforming in any environmental test	
		Acceptance number	Rejection number
1 to 100	1	0	1
101 to 300	2	0	1
301 to 500	4	1	2
501 or more	8	2	3

4.4.3.1.1 Rejection. Each of the sample units selected in accordance with table I shall be subjected to the tests of 4.4.3. The results of each test shall be compared with the requirements of this specification. Failure to conform to this specification for any such test shall be counted as a defect, and the unit shall be rejected. If the number of rejections in any sample exceed the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4.3.1.2 Noncompliance. If a sample unit fails environmental inspections, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of products which were manufactured with the same conditions, materials, and processes. If the Government inspector does not consider that the corrective action will enable the product

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to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3.1.3 Reinspection of conforming environmental inspection sample units. Unless otherwise specified, sample units which have been subjected to and passed environmental inspections may be accepted on contract, provided that they are resubjected to and pass production inspection after repair of all visible damage (see 6.1.1).

4.4.4 Reliability demonstration. Reliability demonstration shall be performed in accordance with MIL-STD-781.

4.4.4.1 Reliability demonstration test plans. The contractor shall submit to the command or agency concerned a reliability demonstration test plan which conforms to the detailed requirements of MIL-STD-781, and the requirements of this specification (see 6.1.2).

4.4.4.2 Reliability qualification tests. Qualification testing shall be performed on at least five production units. Testing shall be performed at test level C with test plan V. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan V.

4.4.4.3 Reliability sampling tests. Reliability sampling tests shall be conducted throughout production to ensure continued production of acceptable equipment. Sampling testing shall be at test level C, with test plan IVa of MIL-STD-781. At least three units shall be randomly selected from each month's production for reliability sampling tests. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan V.

4.4.4.4 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth herein.

4.4.4.5 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer designs, corrective action shall be performed on all equipment delivered under the contract.

4.4.4.6 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of 4.4.1 to ensure that it meets the requirements of this specification. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it meets all terms and conditions of the contract for acceptance.

4.4.4.7 Reliability test procedures. The reliability tests shall be conducted using the procedures, schedules, and performance limits specified herein as a minimum.

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4.4.4.8 Burn-in. The contractor may elect to operate the reliability test samples for a certain time prior to the initiation of reliability testing. Any such burn-in applied to test samples must be applied to all items of equipment submitted for acceptance. Failures occurring during this period may be considered non-relevant, but shall be recorded and reported. Burn-in time for all items of equipment submitted for acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the test samples.

4.5 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with section 5. First article packs which contain Level A packages shall be subjected to rough handling tests (see 5.1.1).

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Drying shall be in accordance with MIL-P-116.

5.1.1.3 Preservative application. None.

5.1.1.4 Unit packaging. Each radiac set shall be individually packaged in accordance with method IIb of MIL-P-116, ensuring compliance with the General paragraph under Methods of Preservation (Unit Protection) and the Physical Protection paragraph therein, and shall meet the rough handling tests of MIL-P-116. Cushioning and dunnage shall be used to prevent physical or mechanical damage.

5.1.1.4.1 Major components. The radiac set shall be packaged in accordance with method IIb of MIL-P-116. Desiccant shall conform to MIL-D-3464, type II or III.

5.1.1.4.2 Technical manuals. Technical manuals shall be packaged in a waterproof, transparent, heat sealed bag, minimum 4-mil thick, and shall not be placed within the barrier used to package any other items.

5.1.2 Level C. Each complete radiac set shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1.1).

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5.2.1 Level A. Each radiac set, packaged as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, V grade, special use requirement, for type 2 load. In lieu of the closure and waterproofing requirements in the appendix to PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, corners, and manufacturer's joint with waterproof tape, 2-inches minimum width, conforming to PPP-T-60, class 1, or PPP-T-76. Reinforcing shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. Each radiac set, packaged as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class weather-resistent. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. Items, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier and which will ensure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

5.3 Special instructions. In addition to the preservation, packaging, and packing specified herein, preparation for delivery procedures for items containing radioactive material shall be accomplished in accordance with MIL-E-17555, and shall ensure compliance with Title 49, Code of Federal Regulations, referenced therein.

5.4 Unitized loads. When applicable, items packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to one destination equal 40 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated in shipping containers specified in 5.2. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard and destinations shall be unitized by hull number.

5.5 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate, and exterior container shall be marked in accordance with MIL-STD-129 (see 6.1.1) and shall ensure compliance with the provisions of MIL-M-19590. Marking shall include set or assembly markings, and special marking for method II packs.

5.6 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified (see 6.1.1), loose-fill polystyrene shall not be used for packaging or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following

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6.1.1 Procurement requirements.

(a) Title, number, and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Finish, if other than corrosion treated or painted gray (see 3.4.8.2.3).
(d) Disposition of sample units (see 4.4.3.1.3).
(e) Levels of preservation, packaging, packing, and applicable marking (see section 5).
(f) Use of loose-fill polystyrene (see 5.6).

6.1.2 Contract data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification includes, but is not restricted to the following:

(a) Preliminary plans and test procedures (see 3.1)
(b) Reliability program plan (see 3.3.2)
(c) Reliability prediction (see 3.3.3)
(d) Reliability report (see 3.3.5)
(e) Contractors quality assurance plan (see 4.1.1.1)
(f) Test procedures (see 4.1.3)
(g) First article inspection report (see 4.3.5)
(h) Reliability test plan (see 4.4.4.1)

6.2 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN, IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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